

Product Requirements Document (PRD): TukTukTrack

1. Product Concept & Vision

Product Name: TukTukTrack

Vision: A comprehensive, real-time logistics and fleet management platform designed to revolutionize how shipments are tracked, managed, and delivered for businesses handling sensitive and time-critical deliveries.

Core Value Proposition: TukTukTrack brings administrators, drivers, and customers together into a single unified platform featuring real-time GPS tracking, IoT-powered cold chain monitoring, powerful analytics, and rigorous role-based access control.

2. Target Audience & User Personas

2.1 The Administrator (Admin)

- **Goal:** Monitor overall logistics operations, manage the fleet, coordinate personnel, and ensure high delivery success rates.
- **Needs:** A dashboard showing system-wide statistics; tools to manage drivers, typical users, and shipments; the ability to assign hardware tracking devices; and capabilities to review analytics and configure system alerts.

2.2 The Driver

- **Goal:** Fulfill deliveries efficiently while ensuring the integrity of sensitive packages.
- **Needs:** A specialized dashboard to view assigned shipments, update delivery statuses, monitor active hardware sensors (such as MPU-6050 and temperature/humidity modules), and easily contact dispatcher support.

2.3 The Customer (User)

- **Goal:** Send and track packages securely and reliably.
 - **Needs:** A simple interface to request shipments, access real-time tracking links, view past shipment history, analyze real-time sensor charts, manage account details, and reach customer service via an AI Chatbot.
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3. Core Features & Requirements

3.1 Real-time GPS Tracking

- **Live Monitoring:** Integration with Leaflet.js / Google Maps API to provide live GPS updates for all active shipment locations.
- **Route Visualization:** Showing estimated time of arrival (ETA) and optimal delivery routes.
- **Geofencing:** Delivery zone tracking with automated movement alerts.

3.2 Role-Based Access Control (RBAC) & Security

- **Authentication:** Firebase Authentication handling robust session management and data isolation (featuring a custom separate logic loop for Admin sessions via `admin-auth.js`).
- **Workflows:** Admin approval required for critical operations and new system registrations.
- **Guards:** Secure route guards ensuring unauthorized users cannot access privileged pages. Enforced Database rules (`firestore.rules` and `database.rules.json`).

3.3 Analytics & Reporting

- **Dashboards:** Performance metrics, operational costs, and key performance indicators (KPIs) available at a glance.
- **Reporting:** Capabilities to export detailed analytics data (PDF/CSV) covering delivery success rates and operational insights.

3.4 Cold Chain & IoT Monitoring

- **Environment Sensor Integration:** Real-time tracking of temperature and humidity (crucial for perishable goods like food and medical supplies).
- **Shock/Tilt Detection:** MPU-6050 sensor data tracking to ensure package handling compliance and track damage incidents.
- **Alerts Configuration:** Automated threshold-based notifications when sensors log values outside of safe limits.

3.5 AI-Powered Customer Support

- **Chatbot Integration:** An intelligent virtual assistant to parse quick questions and solve minor support inquiries efficiently.
- **Escalation Protocol:** Seamless transition from the bot to human customer support handlers utilizing Firebase for ticket administration.

3.6 Progressive Web App (PWA) Capabilities

- **Responsive Layout:** A mobile-first paradigm designed with Tailwind CSS v3.4 ensuring pixel-perfect operation on any device.

- **Installability:** App manifest support allowing the platform to be saved to a user's device for near-native responsiveness.
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4. Technical Architecture

4.1 Frontend Infrastructure

- **Languages:** HTML5, CSS3, JavaScript (ES6+).
- **Styling:** Tailwind CSS v3.4 (compiled via PostCSS in watch mode) + Lucide Icons for vector-based graphics.
- **Mapping Engine:** Leaflet.js / Google Maps API integrations.

4.2 Backend & Cloud Infrastructure Framework

- **Auth:** Firebase Authentication (Handles Phone, Email, and Recaptcha processes).
 - **Database (Primary):** Cloud Firestore (NoSQL document storage optimized for users, shipments, and roles).
 - **Database (IoT Stream):** Firebase Realtime Database (Optimized for high-frequency low-latency updates emitted by GPS modules and environment sensors).
 - **Storage:** Firebase Cloud Storage (Handling unstructured data such as user avatars and document images).
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5. System Structure & Directory Breakdown

Main Access Endpoints

- **Public Space:** `index.html` (Landing), `login.html`, `register.html`, `track.html`
- **Admin Space:** `admindashboard.html`, `admin-users.html`, `admin-drivers.html`, `admin-shipments.html`, `admin-tracking.html`, `admin-alerts.html`, `admin-assign-device.html`
- **Driver Space:** `driver-dashboard.html`, `driver-sensor-status.html`, `contact-service-driver.html`
- **User Space:** `dashboard.html`, `request.html`, `profile.html`, `alerts.html`, `sensor-status.html`, `contact-service-user.html`

JavaScript Modular Ecosystem (/js)

- `auth.js` / `admin-auth.js` / `auth-ui.js`: Primary authentication, token handling, and UI state switches based on RBAC.
- `main.js`: Core shared application logic encompassing layout interactions.
- `chatbot.js`: Embedding AI conversational elements and support triggers.

- **sensor-status.js:** Reactive listeners querying the Firebase Real-time Database to manipulate front-end sensor dashboard UI states (charts/graphs).
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6. Future Scope & Enhancements

- **Predictive AI Routing:** Applying Machine Learning models against historical traffic data to dynamically optimize routes.
- **Hardware API Extensions:** Deeper integration options supporting external ESP32/Arduino firmwares explicitly built for TukTukTrack payloads.
- **Third-Party Logistics (3PL) Adapters:** API bridges built to seamlessly hand off shipments to external couriers for last-mile delivery tracking.